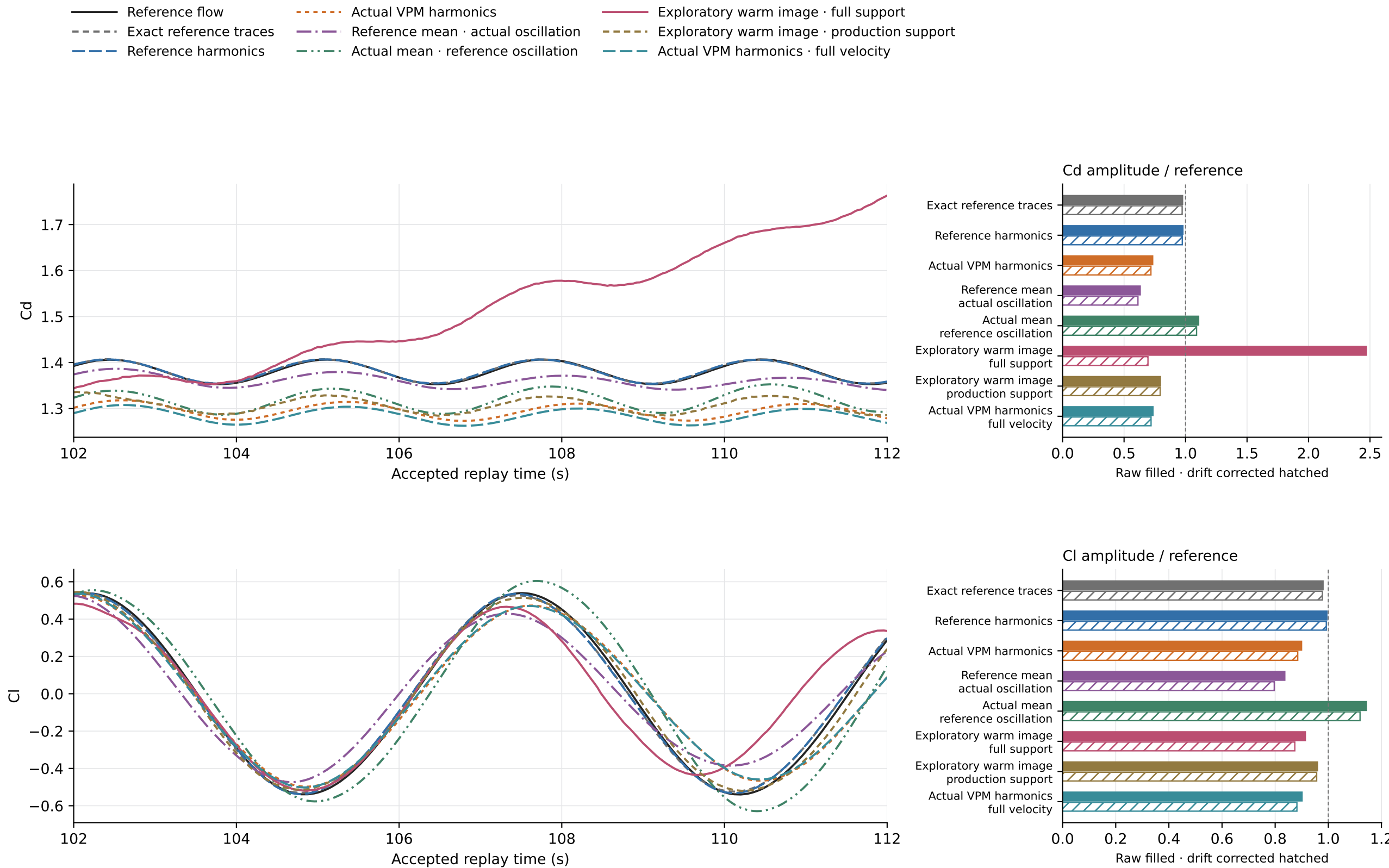


Cylinder boundary-input force controls

Measured 102–112 s: complete cycles after 2 s from the common mapped reference initialization



Input/control	Cd raw	Cd drift	Cl raw	Cl drift	Cd/Cl cycles
Reference flow	0.053701	0.053699	1.078737	1.078404	3/1
Exact reference traces	0.052400	0.052221	1.057401	1.054364	3/1
Reference harmonics	0.052539	0.052273	1.072816	1.070331	3/1
Actual VPM harmonics	0.039347	0.038578	0.970176	0.954368	2/1
Reference mean · actual oscillation	0.033905	0.032918	0.902104	0.858965	2/1
Actual mean · reference oscillation	0.059419	0.058517	1.233346	1.207189	3/1
Exploratory warm image · full support	0.132682	0.037289	0.985590	0.942445	1/1
Exploratory warm image · production support	0.042713	0.042609	1.034673	1.031166	3/1
Actual VPM harmonics · full velocity	0.039458	0.038639	0.972147	0.951034	2/1

Forced FVM controls. No phase or amplitude rescaling; source clocks/field origins are recorded separately.

A 10 s response and one complete lift cycle do not establish autonomous coupled force recovery.

Warm image controls contain growing auxiliary circulation; they cannot establish a finite-wake cause.